Bridging the Thermal Gap: Data-Model Collaborative Transfer for Infrared Perception

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***Abstract*—Infrared visual perception fundamentally differs from visible-domain vision due to its distinct thermal imaging mechanism and varying imaging conditions. Existing methods typically decouple data construction from model transfer, overlooking their interdependence: model accuracy and robustness depend critically on the coverage and composition of training observations, while effective model transfer requires explicit modeling of infrared-specific imaging characteristics. To address this issue, we propose a data-model collaborative transfer framework for infrared visual perception. At the data level, a task-responsive generator learns weights over a feasible infrared imaging-variation space to construct informative training observations. At the model level, a formation-guided representation module derives complementary radiance-structure and response-detail descriptors from infrared observations. We further design a thermal-aware adapter that uses these descriptors to modulate intermediate representations and produce lightweight residual corrections for model transfer. A single-loop training strategy coordinates data generation, representation learning, and adapter optimization without expensive nested optimization. Extensive experiments on image enhancement, depth estimation, small-target detection, and semantic segmentation demonstrate consistent performance improvements across multiple foundation-model backbones with a small trainable parameter budget. Code is available at https://github.com/QilongZhao-86/DMCT_InfraredVision.**

***Index Terms*—Infrared visual perception, data-model collaborative learning, unified transfer framework.**

I. Introduction

Infrared visual perception [1]–[3] is fundamental to robust scene understanding, as it provides complementary information in conditions where visible-light sensors can degrade severely. Unlike visible imaging, which primarily depends on reflected illumination, infrared sensors capture thermal radiation emitted by objects. This imaging mechanism makes infrared observations less sensitive to illumination changes and often more reliable in degraded-visibility environments, such as smoke, haze, and darkness [4], [5]. These properties have enabled a wide range of applications, including autonomous driving, surveillance, and search-and-rescue operations [6].

In recent years, infrared visual perception has advanced along two major directions: visual enhancement and semantic perception, such as small-target detection and segmentation. Specifically, infrared enhancement aims to recover informative content from degraded observations, where corruptions often arise from sensor artifacts and environmental imaging factors, including fixed-pattern/non-uniformity noise, blur, and low resolution [7], [8]. Existing learning-based methods usually handle infrared degradation by designing specialized networks for specific corruption patterns. For stripe or non-uniformity noise removal, residual multi-scale networks and attention-based architectures [8], [9] have been widely explored. For super-resolution, recent methods exploit lightweight attention [10] and dynamic filtering [11] to recover weak textures. Despite these advances, a persistent bottleneck lies in the limited coverage of infrared imaging conditions, since sufficient paired infrared data are often scarce and difficult to collect.

Parallel to enhancement, infrared semantic perception, such as small-target detection, faces the challenge of extracting salient objects from clutter-dominant backgrounds under low signal-to-noise ratios (SNRs) and extreme target imbalance [12], [13]. Recent advances have shifted toward task-driven deep networks that enhance target saliency while suppressing background interference, commonly through multi-scale feature aggregation [6], attention-based interaction [14], and frequency-aware modeling [15]. Feature-level modules further improve detection ability under weak thermal contrast, for example, by leveraging edge-aware cues [16] or spatial-channel interactions [17]. Despite steady progress, infrared perception still faces two core challenges: dynamic imaging variations caused by sensor, optics, and environmental factors, and thermal structures that are fundamentally different from those in visible-domain imagery. These characteristics make models trained on fixed datasets brittle under changing infrared scenarios, motivating a collaborative framework that jointly considers data construction and model transfer.

The recent emergence of Visual Foundation Models (VFMs) [18]–[20] provides a promising paradigm for infrared foundation-model fine-tuning. Generative VFMs, such as Stable Diffusion [18], encode rich structural priors that can benefit infrared restoration, while promptable and self-supervised VFMs, such as SAM [19] and DINO [20], provide strong transferable representations for dense perception. Consequently, adapting these visible-pretrained models to infrared tasks through Parameter-Efficient Fine-Tuning (PEFT) strategies, such as Low-Rank Adaptation (LoRA) [21]–[23] and adapters [24]–[26], has become an attractive direction. Nevertheless, a critical challenge remains: the visual priors learned from visible imagery are not naturally aligned with thermal infrared observations. As a result, generic lightweight

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updating schemes often produce limited or inconsistent gains in infrared settings, because their updates cannot adaptively preserve the inhere characteristics of infrared imaging.

To address these challenges, we propose a data-model collaborative transfer framework for infrared visual perception[1]. The framework couples data construction with model transfer under infrared imaging priors. At the data level, it learns task-responsive weights over a feasible infrared imaging-variation space to construct informative training observations. At the model level, a formation-guided representation module derives complementary radiance-structure and response-detail descriptors. These descriptors condition lightweight adapters through feature-wise modulation, enabling thermal-aware model transfer. To coordinate these coupled components, we further develop a single-loop training strategy that stabilizes collaborative optimization without requiring inner-loop convergence. Experiments on four perception tasks demonstrate consistent performance promotion while maintaining a small parameter budget. Our contributions are summarized as follows:

- To the best of our knowledge, this is the first systematic framework to investigate the transfer of visual foundation models to infrared perception from a data-model collaborative perspective.
- We develop a task-responsive data generation strategy that learns weights over a feasible infrared imaging-variation space, enabling the construction of informative training observations under diverse imaging conditions.
- We propose a thermal-aware model transfer mechanism that uses formation-guided radiance-structure and response-detail descriptors to condition lightweight adapters, improving the alignment between foundation-model representations and infrared observations.
- We design a single-loop training strategy to coordinate task-responsive data generation, formation-guided representation learning, and lightweight model transfer without requiring inner-loop convergence.

This work substantially extends our preliminary conference version DEAL [7]. While DEAL focuses on data-adaptive adversarial learning for infrared image enhancement, the present work addresses the broader problem of infrared visual perception through a data-model collaborative transfer framework. The extension is threefold. First, the degradation-oriented data generation in DEAL is generalized into task-responsive data generation over a feasible infrared imaging-variation space, enabling informative observation synthesis across different perception tasks. Second, we introduce a formation-guided model-transfer mechanism, where radiance-structure and response-detail descriptors condition lightweight adapters for infrared-specific representation transfer. Third, we extend the study from a single enhancement task to multiple foundation backbones and representative infrared perception tasks, including image enhancement, depth estimation, small-target detection, and semantic segmentation.

[1]A preliminary version of this work has been published in [7].

II. Related Work

A. Infrared Visual Perception

Infrared visual perception can be roughly divided into two categories, including visual enhancement and infrared semantic understanding. Recently, a wave of infrared image restoration methods has emerged. Addressing the lack of texture, some researchers introduced CoRPLE [27], leveraging contourlet transforms and prompt learning to decouple high-frequency details from backgrounds. Liu et al. proposed the Progressive Prompt Fusion Network (PPFN), which treats degradation removal as a sequential prompt-guided process to handle complex composite corruptions [10]. Generative paradigms have also been adapted for infrared imaging; for instance, Li et al. proposed Difiisr [11], a diffusion model explicitly tailored for infrared super-resolution. By incorporating a gradient-guided auxiliary branch, Difiisr effectively constrains the diffusion process to preserve sharp thermal structures while hallucinating realistic details. Similarly, tackling data efficiency, Liu et al. introduced DEAL, which formulates a dynamic adversarial game where a generator simulates thermal-specific degradations to challenge the restoration network, demonstrating that learning from adaptive synthesized examples can significantly improve restoration quality [7]. Chen et al. model real-world thermal degradation under background thermal noise and streak interference for more realistic super-resolution settings [28]. Additionally, joint formulations that couple super-resolution with correction reflect the practical need to handle multiple corruption sources simultaneously [29]. Most methods are task-specific and do not provide a unified mechanism to improve robustness.

For infrared semantic perception, Infrared Small Target Detection (IRSTD) has been extensively studied due to its practical importance and challenging target-background imbalance. Dai et al. pioneered ACM, introducing point-wise channel attention to exchange low-level and high-level semantics [30]. Addressing the issue of target submersion in deep layers, Li et al. developed DNA-Net, employing densely nested interaction modules to maintain the semantic integrity of tiny targets during repetitive downsampling [6]. Wu et al. further advanced this with UIU-Net, embedding a tiny U-Net within a larger backbone to handle extreme scale variations [31]. More recent studies continue to explore frequency-aware [16], [17] enhancements or progressive refinement [32], [33] for suppressing textured clutter and improving detection ability, especially for moving/dim targets. Despite the steady progress, existing restoration and perception methods still rely on limited paired supervision and few investigate the imaging-consistent representation that separates stable thermal structure from corruption. These limitations motivate frameworks that jointly consider data construction and model transfer, so that infrared perception can be improved coherently under scarce supervision and dynamic imaging conditions.

B. Data Adaptation for Robust Vision

Initially, static methods like Mixup [34] and CutMix [35] improved robustness via linear interpolation with diverse augmentation operations. This line of research was further

extended by automated search strategies, such as AutoAugment [36] and RandAugment [37], which utilize reinforcement learning to optimize augmentation policies from a discrete search space. Most recently, Madry et al. established the theoretical foundation with PGD adversarial training [38], generating perturbations that maximize loss to enforce resistance against worst-case attacks. Building on this, methods like Adversarial AutoAugment [39] and DeepAugment [40] train generator networks to produce hard augmentations that challenge the perception model. Lastly, some recent works also explore generative models to synthesize realistic weather or sensor degradations [41], [42]. However, these methods are largely developed for visible imagery and remain modality-agnostic: they rely on generic transformations without explicitly modeling infrared imaging regularities or supporting collaborative model transfer.

C. Parameter-Efficient Transfer of Foundation Models

Parameter-Efficient Fine-Tuning (PEFT) has emerged as an effective paradigm for transferring foundation models to specific domains. Existing PEFT methods can be broadly categorized into three representative families: adapter-based approaches [24]–[26], which insert lightweight trainable modules into the network; low-rank adaptation [21]–[23], which learns low-rank factors to approximate weight updates; and prompt-based approaches [43]–[45], which inject learnable tokens into the model inputs to steer adaptation. Adapter-based schemes, in particular, have achieved notable progress in dense vision tasks. For instance, ViT-Adapter [24] introduces spatial priors and multi-scale feature interactions for dense prediction. Earth-Adapter [46] combines frequency decomposition with a mixture of adapters to mitigate artifacts in remote sensing features. PointGST [47] performs adaptation in the graph spectral domain, reducing token confusion while incorporating the intrinsic geometry of downstream point clouds. Rein [48] uses trainable tokens to refine intermediate features at the instance level for domain-generalized semantic segmentation. These approaches provide effective mechanisms for adapting model representations. Yin et al. proposed Mona [25], leveraging vision-friendly filters and modulation mechanisms to enhance adaptation for detection and segmentation. Chen et al. proposed RSPrompter, which generates instance-specific prompts to align aerial images with the object-centric priors of CLIP [26]. However, applying visible-pretrained foundation models to infrared perception faces a unique physical modality gap, the difference between reflective illumination and thermal radiation. This motivates a collaborative framework that couples imaging-driven data construction with thermal-aware model transfer. Specifically, the generator learns task-responsive weights over a constrained infrared imaging-variation space by maximizing the current training loss, while the formation-guided representation and adapter are updated to improve task performance.

III. The Proposed Method

A. Collaborative Transfer Framework

Although visual foundation models provide strong general-purpose representations, transferring them to infrared perception remains challenging because of the discrepancy between visible-domain priors and thermal observations. From the data perspective, fixed infrared training sets cannot adequately cover diverse and dynamically changing conditions. From the model perspective, generic modules are largely modality-agnostic and therefore struggle to preserve radiance-dominant structures while compensating response-sensitive details.

To address these challenges, we propose a data-model collaborative transfer framework for infrared visual perception. The framework consists of three coupled components: a task-responsive infrared data generator $\mathcal{G}(\boldsymbol{\varphi})$ parameterized by $\boldsymbol{\varphi}$, a formation-guided representation module $\mathcal{P}(\boldsymbol{\theta})$ with parameters $\boldsymbol{\theta}$, and a thermal-aware adapter $\mathcal{A}(\boldsymbol{\omega})$ with parameters $\boldsymbol{\omega}$. Given an infrared input $\mathbf{x}$, $\mathcal{G}(\boldsymbol{\varphi})$ first models task-responsive weights over a feasible infrared imaging-variation space and constructs an informative observation: $\tilde{\mathbf{x}} = \mathcal{G}(\mathbf{x};\boldsymbol{\varphi})$. The representation module $\mathcal{P}(\boldsymbol{\theta})$ then extracts a formation-guided descriptor: $\mathbf{p} = \mathcal{P}(\tilde{\mathbf{x}};\boldsymbol{\theta})$, where $\mathbf{p}$ encodes radiance-dominant structural cues and response-sensitive detail cues. Finally, $\mathbf{p}$ conditions the lightweight adapter $\mathcal{A}(\boldsymbol{\omega})$ to guide infrared-aware model transfer while keeping the foundation-model backbone frozen. The learning objective is formulated as:

$$\min_{\boldsymbol{\theta}} \quad \mathcal{L}_{\mathrm{val}}\Big(\boldsymbol{\theta},\, \boldsymbol{\omega}^{\star}(\boldsymbol{\theta},\boldsymbol{\varphi}^{\star});\, \boldsymbol{\varphi}^{\star}\Big) \tag{1}$$

$$\text{s.t.} \quad \begin{cases} \boldsymbol{\omega}^{\star}(\boldsymbol{\theta},\boldsymbol{\varphi}) = \arg\min\limits_{\boldsymbol{\omega}} \mathcal{L}_{\mathrm{tr}}(\boldsymbol{\omega},\boldsymbol{\theta};\boldsymbol{\varphi}), \\ \boldsymbol{\varphi}^{\star}(\boldsymbol{\omega},\boldsymbol{\theta}) = \arg\max\limits_{\boldsymbol{\varphi}\in\mathcal{C}} \mathcal{L}_{\mathrm{tr}}(\boldsymbol{\omega},\boldsymbol{\theta};\boldsymbol{\varphi}). \end{cases} \tag{2}$$

This formulation defines a nested optimization problem that jointly coordinates data construction, formation-guided representation learning, and lightweight model transfer. In Eq. (2), the data parameters $\boldsymbol{\varphi}$ identify task-informative imaging variations within the feasible set $\mathcal{C}$ and dynamically construct training observations, while the adapter parameters $\boldsymbol{\omega}$ minimize the task loss $\mathcal{L}_{\mathrm{tr}}$ under the current data configuration. At the outer level in Eq. (1), the representation parameters $\boldsymbol{\theta}$are optimized by minimizing $\mathcal{L}_{\mathrm{val}}$, encouraging the learned radiance-structure and response-detail descriptors to provide effective conditioning cues for model transfer. In this way, data construction and model transfer are collaboratively optimized under infrared imaging priors.

B. Task-Responsive Data Generation

Infrared observations exhibit imaging variations that differ from visible-domain imagery and often co-occur in practical sensing environments. We consider five representative variations: contrast response caused by weak temperature differences and environmental radiation, spatial blur introduced by motion and defocus, sensor response involving fixed-pattern/stripe noise and optical noise, and sampling-resolution variation caused by hardware constraints, as illustrated in Fig. 1(a). Together, these variations define a feasible infrared imaging-variation space for constructing diverse and informative training observations. We parameterize the data generator $\mathcal{G}(\boldsymbol{\varphi})$ as a differentiable policy network. Instead of using fixed or randomly sampled variation parameters, $\mathcal{G}(\boldsymbol{\varphi})$ predicts task-

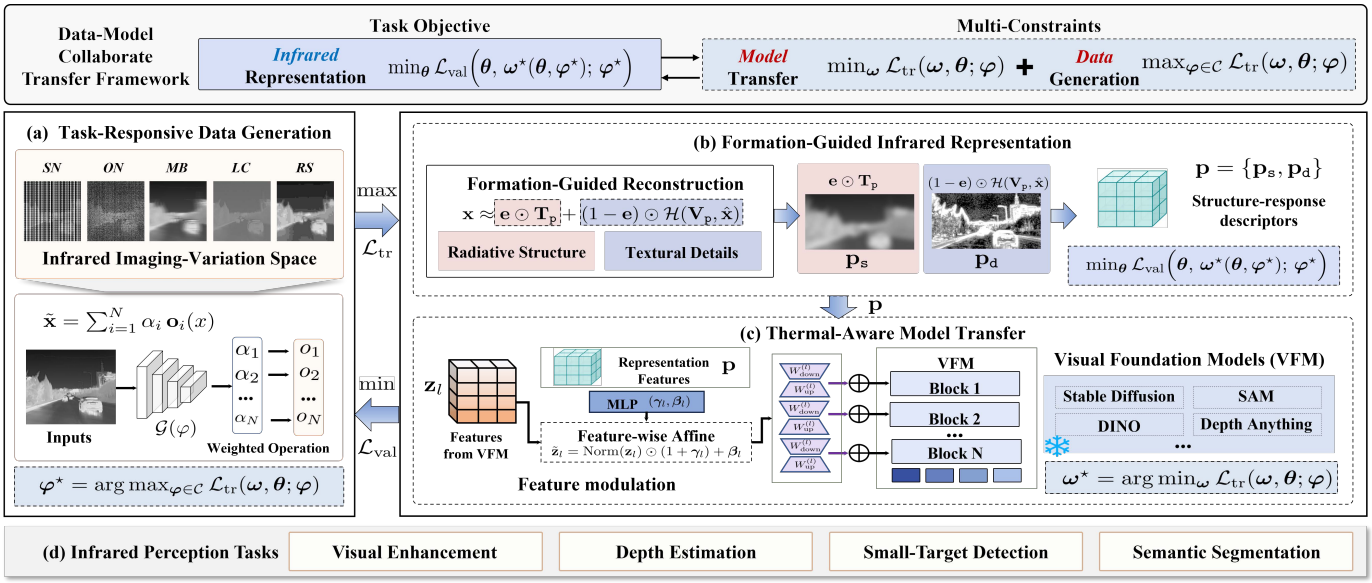


Fig. 1. Overview of the proposed data-model collaborative transfer framework. (a) Task-driven data generation learns task-responsive weights over a feasible infrared imaging-variation space, including Stripe Noise (SN), Optical Noise (ON), Motion Blur (MB), Low Contrast (LC), and Resolution Variation (RS). (b) Formation-guided infrared representation derives radiance-structure and response-detail descriptors from infrared inputs. (c) Thermal-aware model transfer uses these descriptors to condition lightweight adapters through feature-wise modulation. (d) The framework is instantiated for four representative infrared perception tasks: image enhancement, depth estimation, small-target detection, and semantic segmentation.

responsive distribution weights $\boldsymbol{\alpha}$ to synthesize informative training observations:

$$\tilde{\mathbf{x}} = \sum_{i=1}^{N} \alpha_i \, \mathbf{o}_i(\mathbf{x}), \quad \text{s.t.} \ \sum_{i=1}^{N} \alpha_i = 1, \ \alpha_i \geq 0, \tag{3}$$

where $\mathbf{o}_i$ represents a differentiable infrared imaging-variation operator. We enforce $\alpha_i \geq 0$ via softmax, and constrain each operator parameter within predefined physically plausible ranges. The predicted weights $\boldsymbol{\alpha}$ define a distribution over candidate imaging variations, allowing $\mathcal{G}(\boldsymbol{\varphi})$ to adaptively adjust the composition of training data according to the current fine-tuning state. This design does not aim to exactly reproduce the full infrared sensor pipeline; instead, it provides a constrained and differentiable variation space for constructing informative observations. Compared with fixed handcrafted transformations, the generator can dynamically emphasize informative infrared data conditions, thereby improving robustness.

C. Formation-Guided Infrared Representation

Infrared image formation differs fundamentally from visible imaging. Directly modeling infrared observations without considering their formation characteristics may entangle radiance-dominant structures with response-sensitive details, thereby weakening robustness under varying imaging conditions. To preserve this formation structure while remaining compatible with foundation-model transfer, we build upon the Temperature–Emissivity–Texture (TeX) factorization [1] and instantiate a learnable formation-guided proxy representation module $\mathcal{P}(\boldsymbol{\theta})$. Rather than explicitly recovering physical quantities, this module employs a learnable proxy parameterization to derive formation-guided descriptors. According to the TeX model, the infrared radiance $\mathbf{x}_\lambda$ at wavelength λ is represented as a combination of thermal emission and reflected environmental radiation:

$$\mathbf{x}_\lambda = \mathbf{e}_\lambda \odot B_\lambda(\mathbf{T}) + (1 - \mathbf{e}_\lambda) \odot \mathbf{X}_\lambda, \tag{4}$$

where $\mathbf{e}_\lambda \in [0, 1]$ denotes the material-dependent emissivity map, $B_\lambda(\mathbf{T})$ represents the blackbody radiance governed by the object's kinetic temperature $\mathbf{T}$, and $\mathbf{X}_\lambda$ denotes the environmental irradiance reflected by the surface. For tractable modeling, TeX locally approximates the environmental term as a linear modulation of global scene statistics, i.e., $\mathbf{X}_\lambda \approx \mathbf{V} \cdot \hat{\mathbf{x}}_\lambda$, where $\hat{\mathbf{x}}_\lambda$ summarizes the global observation and $\mathbf{V}$ denotes pixel-wise texture coefficients [49]. Our goal is not to recover physical temperature, emissivity, or environmental irradiance, but to derive formation-guided proxy descriptors that provide infrared-specific conditioning cues for thermal-aware model transfer. As shown in Fig. 1(b), we implement $\mathcal{P}(\boldsymbol{\theta})$ using a learnable low-/high-frequency decomposition. This decomposition serves as a practical proxy for separating spatially coherent structures from residual details [50], [51]. Formally, given an infrared input $\mathbf{x}$, we perform the following learnable decomposition: $\mathbf{T}_\mathrm{p} = \mathcal{F}_\mathrm{low}(\mathbf{x}), \quad \mathbf{V}_\mathrm{p} = \mathbf{x} - \mathcal{F}_\mathrm{low}(\mathbf{x})$, where $\mathcal{F}_\mathrm{low}(\cdot)$ denotes a learnable guided-filter operator [52], [53]. We regard $\mathbf{T}_\mathrm{p}$ as a proxy for radiance-dominant structures and $\mathbf{V}_\mathrm{p}$ as a proxy for response-sensitive details. Based on these components, we introduce an emissivity-inspired spatial proxy $\mathbf{e}_\mathrm{p} = g_s(\mathbf{x})$ through a spatial attention mechanism, which assigns spatially varying weights to structure-dominant responses. Meanwhile, a channel attention module produces a global context descriptor $\hat{\mathbf{x}} = g_c(\mathbf{x})$ to summarize scene-level response information. These components yield the following formation-guided proxy composition:

$$\mathbf{x} \approx \mathbf{e}_\mathrm{p} \odot \mathbf{T}_\mathrm{p} + (1 - \mathbf{e}_\mathrm{p}) \odot \mathcal{H}(\mathbf{V}_\mathrm{p}, \hat{\mathbf{x}}), \tag{5}$$

where $\mathcal{H}(\cdot)$ transforms the response-sensitive detail component conditioned on the global context descriptor $\hat{\mathbf{x}}$. The parameter set $\boldsymbol{\theta}$ includes all learnable parameters in $\mathcal{F}_{\text{low}}$, g_c, g_s, and $\mathcal{H}$. Finally, the resulting proxy components are collected into $\mathbf{p}$ and used as infrared-specific conditioning cues for lightweight adapters, thereby enabling formation-guided model transfer.

D. Thermal-Aware Model Transfer

Adapters support parameter-efficient model transfer by introducing lightweight residual branches with a small number of trainable parameters while keeping the backbone frozen. However, most existing adapters are modality-agnostic and are not explicitly conditioned on the formation characteristics of infrared observations. Consequently, they may inadequately preserve structure-dominant information or compensate response-sensitive details. To address this limitation, we propose a thermal-aware adapter that leverages formation-guided descriptors to steer model transfer for infrared perception. Concretely, we construct two complementary factors, i.e., $\mathbf{p} = \{\mathbf{p}_{\mathtt{s}}, \mathbf{p}_{\mathtt{d}}\}$, where $\mathbf{p}_{\mathtt{s}} = \mathbf{e}_{\mathtt{p}} \odot \mathbf{T}_{\mathtt{p}}$ and $\mathbf{p}_{\mathtt{d}} = (1 - \mathbf{e}_{\mathtt{p}}) \odot \mathcal{H}(\mathbf{V}_{\mathtt{p}}, \hat{\mathbf{x}})$. The structure factor $\mathbf{p}_{\mathtt{s}}$ emphasizes spatially coherent and radiance-dominant content, whereas the detail factor $\mathbf{p}_{\mathtt{d}}$ captures response-sensitive information associated with sensor responses and imaging variations. The two factors are embedded into guidance features and incorporated into a lightweight residual adapter, as illustrated in Fig. 1(c).

To encode their complementary characteristics, we first transform the structure and detail factors into frequency-enhanced guidance representations. Specifically, the two factors are processed using FFT-based frequency enhancement to capture complementary structural and detail cues. The resulting representations are then aggregated and fused with the projected backbone feature to form a compact guidance descriptor. Let $\mathbf{z}_l$ denote the intermediate feature at the l-th layer, and let $\mathbf{g}_l$ denote the guidance descriptor obtained by combining the frequency-enhanced structure and detail representations with the projected intermediate feature. A lightweight MLP maps $\mathbf{g}_l$ to the modulation parameters $(\boldsymbol{\gamma}_l, \boldsymbol{\beta}_l)$, which perform feature-wise affine modulation:

$$\tilde{\mathbf{z}}_l = \mathrm{Norm}(\mathbf{z}_l) \odot (1 + \boldsymbol{\gamma}_l) + \boldsymbol{\beta}_l. \tag{6}$$

Here, $\boldsymbol{\gamma}_l$ and $\boldsymbol{\beta}_l$ are broadcast along the spatial or token dimensions. The resulting modulation selectively rescales and shifts the intermediate representation according to the formation-guided infrared descriptors. The modulated feature is subsequently passed through a lightweight residual bottleneck to produce a residual correction:

$$\Delta\mathbf{z}_l = W_{\text{up}}^{(l)} \sigma\Big(W_{\text{down}}^{(l)} \tilde{\mathbf{z}}_l\Big), \quad \mathbf{z}_l' = \mathbf{z}_l + \Delta\mathbf{z}_l. \tag{7}$$

where $W_{\text{down}}^{(l)}$ and $W_{\text{up}}^{(l)}$ denote the bottleneck projections, and $\sigma(\cdot)$ denotes a nonlinear activation function. In this way, the proposed adapter connects formation-guided infrared representation with lightweight model transfer. The descriptors provide infrared-specific conditioning cues, while the residual branch converts these cues into feature-wise modulation and representation correction.

E. Single-Loop Collaborative Optimization

Optimizing the bilevel objective in Eq. (1) is challenging because it couples two mutually dependent yet directionally opposed processes: task-responsive data generation and thermal-aware model transfer. The generator seeks informative imaging variations by maximizing the training loss, whereas the representation module and lightweight adapter aim to improve perception under the generated observations. Direct nested optimization requires repeated inner-loop convergence and may become unstable when the data distribution changes faster than the model can respond. To address this issue, we develop a single-loop collaborative optimization strategy that updates the data generator, formation-guided representation module, and lightweight adapter within each iteration, without requiring inner-loop convergence. Rather than enforcing exact lower-level optimality, we encourage the adapter parameters to satisfy the first-order stationarity condition of the lower-level objective. Specifically, we define the stationarity residual as

$$h(\boldsymbol{\theta}, \boldsymbol{\omega}; \boldsymbol{\varphi}) = \|\nabla_{\boldsymbol{\omega}} \mathcal{L}_{\text{tr}}(\boldsymbol{\omega}, \boldsymbol{\theta}; \boldsymbol{\varphi})\|_2^2 .$$

When $h = 0$, $\boldsymbol{\omega}$ satisfies the first-order stationarity condition of the lower-level objective. We then construct the following dynamic-penalty objective:

$$\Phi_k(\boldsymbol{\theta}, \boldsymbol{\omega}; \boldsymbol{\varphi}) = \mathcal{L}_{\text{val}}(\boldsymbol{\theta}, \boldsymbol{\omega}; \boldsymbol{\varphi}) + \lambda_k h(\boldsymbol{\theta}, \boldsymbol{\omega}; \boldsymbol{\varphi}), \tag{8}$$

where λ_k balances the gradient contributions of the validation objective and the stationarity penalty. Specifically, we adopt the following gradient-scale balancing rule [54]:

$$\lambda_k = \beta \, \frac{\big\|\nabla_{(\boldsymbol{\theta},\boldsymbol{\omega})} \mathcal{L}_{\text{val}}(\boldsymbol{\theta}_k, \boldsymbol{\omega}_k; \boldsymbol{\varphi}_{k+1})\big\|_2}{\big\|\nabla_{(\boldsymbol{\theta},\boldsymbol{\omega})} h(\boldsymbol{\theta}_k, \boldsymbol{\omega}_k; \boldsymbol{\varphi}_{k+1})\big\|_2 + \delta}, \tag{9}$$

where β is a scaling coefficient and $\delta > 0$ ensures numerical stability. This rule keeps the stationarity penalty commensurate with the validation objective, preventing either term from dominating the optimization.

At iteration k, given $(\boldsymbol{\theta}_k, \boldsymbol{\omega}_k, \boldsymbol{\varphi}_k)$, we first update the infrared data generator through one projected gradient-ascent step:

$$\boldsymbol{\varphi}_{k+1} \leftarrow \mathrm{Proj}_{\mathcal{C}} \left(\boldsymbol{\varphi}_k + \eta_{\boldsymbol{\varphi}} \nabla_{\boldsymbol{\varphi}} \mathcal{L}_{\text{tr}}(\boldsymbol{\omega}_k, \boldsymbol{\theta}_k; \boldsymbol{\varphi}_k)\right), \tag{10}$$

where $\mathrm{Proj}_{\mathcal{C}}(\cdot)$ constrains the generator parameters within the feasible infrared imaging-variation space. This update encourages the generator to emphasize informative imaging variations under the current perception model.

Given the updated generator $\boldsymbol{\varphi}_{k+1}$, we jointly update the representation and adapter parameters $(\boldsymbol{\theta}, \boldsymbol{\omega})$ through one gradient-descent step on the dynamic-penalty objective:

$$(\boldsymbol{\theta}_{k+1}, \boldsymbol{\omega}_{k+1}) \leftarrow (\boldsymbol{\theta}_k, \boldsymbol{\omega}_k) - \eta_{(\boldsymbol{\theta},\boldsymbol{\omega})} \nabla_{(\boldsymbol{\theta},\boldsymbol{\omega})} \Phi_k(\boldsymbol{\theta}_k, \boldsymbol{\omega}_k; \boldsymbol{\varphi}_{k+1}), \tag{11}$$

whose gradient can be expanded as

$$\nabla_{(\boldsymbol{\theta},\boldsymbol{\omega})} \mathcal{L}_{\text{val}}(\boldsymbol{\theta}_k, \boldsymbol{\omega}_k; \boldsymbol{\varphi}_{k+1}) + \lambda_k \nabla_{(\boldsymbol{\theta},\boldsymbol{\omega})} h(\boldsymbol{\theta}_k, \boldsymbol{\omega}_k; \boldsymbol{\varphi}_{k+1}). \tag{12}$$

The first term improves validation performance under the current generated observations, while the second reduces the violation of the lower-level stationarity condition. Their joint optimization maintains consistency between formation-guided representation learning and the current model-transfer state.

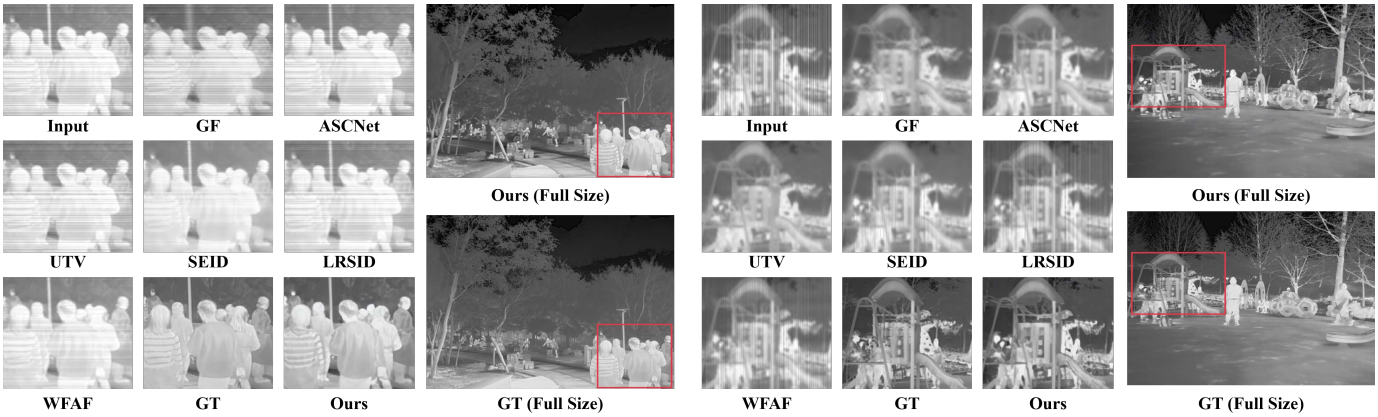


Fig. 2. Qualitative comparison of infrared stripe-noise removal on the HM-TIR dataset.

TABLE I
QUANTITATIVE COMPARISON OF IMAGE RESTORATION PERFORMANCE UNDER MODERATE AND SEVERE STRIPE DEGRADATIONS.

Degrees	Moderate					Severe				
Methods	LPIPS↓	MUSIQ↑	CL-IQA↑	MANIQA↑	FID↓	LPIPS↓	MUSIQ↑	CL-IQA↑	MANIQA↑	FID↓
UTV	0.5849	23.80	0.1280	0.2175	193.5	0.7676	29.05	0.2119	**0.3172**	305.5
WFAF	0.4611	28.07	0.09580	0.1733	166.6	0.6876	30.83	0.1720	0.2386	256.4
GF	0.5485	23.25	0.1165	0.2116	175.0	0.7537	23.74	0.1672	0.2864	284.4
LRSID	0.5391	24.50	0.1278	0.2048	195.1	0.7586	31.57	0.2187	0.3123	304.7
SEID	0.4605	31.25	0.1329	0.1862	141.0	0.6640	40.39	0.2021	0.2915	250.6
ASCNet	0.5103	24.13	0.1178	0.1963	138.1	0.7167	24.15	0.1586	0.2633	238.8
PPFN	0.3188	30.92	0.1195	0.2042	97.16	0.5255	23.15	0.09450	0.1667	190.0
Ours	**0.1238**	**49.66**	**0.2491**	**0.2657**	**43.01**	**0.2020**	**49.46**	**0.2512**	0.2745	**70.81**

Directly evaluating ∇h requires computationally expensive Hessian–vector products (HVPs). We therefore approximate the required HVPs using finite differences of first-order gradients. Let $\mathbf{v}_k = \nabla_{\boldsymbol{\omega}} \mathcal{L}_{\mathrm{tr}}(\boldsymbol{\omega}_k, \boldsymbol{\theta}_k; \boldsymbol{\varphi}_{k+1})$. The gradients of the stationarity residual can be expressed as $\nabla_{\boldsymbol{\omega}} h = 2\nabla^2_{\boldsymbol{\omega}\boldsymbol{\omega}} \mathcal{L}_{\mathrm{tr}} \mathbf{v}_k, \quad \nabla_{\boldsymbol{\theta}} h = 2\nabla^2_{\boldsymbol{\theta}\boldsymbol{\omega}} \mathcal{L}_{\mathrm{tr}} \mathbf{v}_k$.

For a small perturbation $\epsilon > 0$, the corresponding HVPs are approximated as

$$\nabla^2_{\boldsymbol{\omega}\boldsymbol{\omega}} \mathcal{L}_{\mathrm{tr}} \mathbf{v}_k \approx \frac{\nabla_{\boldsymbol{\omega}} \mathcal{L}_{\mathrm{tr}}(\boldsymbol{\omega}_k + \epsilon \mathbf{v}_k, \boldsymbol{\theta}_k) - \nabla_{\boldsymbol{\omega}} \mathcal{L}_{\mathrm{tr}}(\boldsymbol{\omega}_k, \boldsymbol{\theta}_k)}{\epsilon}, \tag{13}$$

and

$$\nabla^2_{\boldsymbol{\theta}\boldsymbol{\omega}} \mathcal{L}_{\mathrm{tr}} \mathbf{v}_k \approx \frac{\nabla_{\boldsymbol{\theta}} \mathcal{L}_{\mathrm{tr}}(\boldsymbol{\omega}_k + \epsilon \mathbf{v}_k, \boldsymbol{\theta}_k) - \nabla_{\boldsymbol{\theta}} \mathcal{L}_{\mathrm{tr}}(\boldsymbol{\omega}_k, \boldsymbol{\theta}_k)}{\epsilon}. \tag{14}$$

For brevity, $\boldsymbol{\varphi}_{k+1}$ is omitted from $\mathcal{L}_{\mathrm{tr}}$ in Eqs. (13) and (14). The finite-difference approximation avoids explicit Hessian construction and requires only two additional first-order gradient evaluations per iteration.

IV. APPLICATIONS

In this section, we describe how the framework is instantiated across four representative infrared perception tasks, including image enhancement, depth estimation, small-target detection, and semantic segmentation, as illustrated in Fig. 1(d).

Infrared Image Enhancement. This task aims to recover high-quality images from degraded observations affected by noise, blur, resolution loss, and sensor-induced artifacts. Compared with visible images, infrared images often exhibit low contrast, weak texture, non-uniform intensity distributions, and structured noise, making faithful restoration particularly challenging. We perform the experiments on a Stable-Diffusion-based restoration model [55] by inserting the proposed adapter into the U-Net, allowing the visible-pretrained diffusion prior to be transferred to infrared image enhancement through lightweight, formation-guided modulation. The evaluation is conducted on the HM-TIR dataset [10], which contains 9,027 training images and 301 testing images with multiple stripe-noise levels and $\times 2/\times 4$ resolution degradations. The model is trained for 10,000 iterations using AdamW with a learning rate of 2×10^{-5}. We report LPIPS to evaluate perceptual similarity, FID to measure distribution discrepancy, and MUSIQ, CL-IQA, and MANIQA to assess no-reference perceptual quality.

Infrared Depth Estimation. This task aims to recover scene geometry from thermal images. In dynamic driving scenarios, motion blur caused by camera motion, platform vibration, or moving objects can weaken structural boundaries, distort object contours, and degrade metric depth prediction. We adopt Depth Anything [56] as the backbone and transfer visible-domain pretrained weights to infrared depth estimation. We use a fixed subset of the MS2 dataset [57], following the same scene-level split for all compared methods, resulting in 3,369 training pairs and 843 testing pairs. We focus on

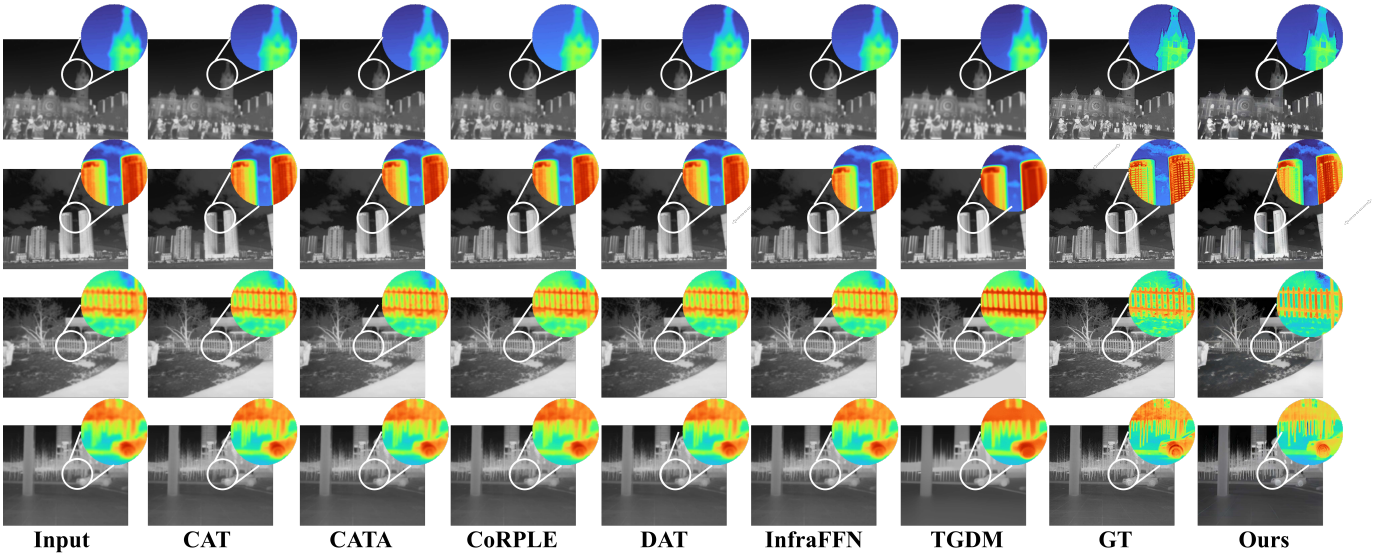


Fig. 3. Qualitative comparison of different methods for image super-resolution at ×2 and ×4 scales.

standardized motion blur to evaluate imaging-robust geometric prediction. For restoration-based baselines, the restored infrared images are fed into the same depth-estimation backbone to ensure a consistent comparison protocol. The model is trained for 30 epochs with a learning rate of 1×10^{-3}. We report standard depth error metrics to evaluate both strict prediction accuracy and global metric consistency.

Infrared Small Target Detection. This task aims to localize tiny targets in cluttered and noisy infrared scenes, where targets often occupy only a few pixels and can be easily confused with background edges, clouds, sea clutter, buildings, or sensor noise. We instantiate our method on SAM2 [58] and evaluate it on NUDT-SIRST [6], IRSTD-1k [16], and IRSTDID-SKY [59]. The input resolution is set to 256×256 for NUDT-SIRST and 512×512 for IRSTD-1k and IRSTDID-SKY. The proposed adapter performs thermal-aware feature modulation, helping the model preserve weak target responses while suppressing background interference under noisy and low-contrast conditions. The model is trained for 300 epochs using Adam with an initial learning rate of 0.01. We report mIoU, F1-score, detection probability P_d, and false alarm rate F_a to evaluate pixel-level segmentation accuracy, target detection capability, and background false-alarm suppression.

Infrared Semantic Segmentation. This task requires pixel-level scene understanding under diverse imaging variations and weak object boundaries. Since infrared images lack rich color and texture cues, objects with similar thermal responses are often difficult to distinguish, requiring robust feature extraction and thermal-aware model transfer. We adopt DINOv2 [20] as the backbone and evaluate robustness under multiple stripe-noise levels and $\times 2/\times 4$ downsampling on the FMB dataset [60]. The model is optimized using AdamW with a learning rate of 1×10^{-4}. The proposed adapter is inserted after transformer blocks, and frequency decomposition is performed in the feature space to derive complementary structure- and detail-sensitive guidance for thermal-aware model transfer. We report class-wise IoU and mean Intersection-over-Union (mIoU) over the semantic categories of FMB.

V. Experimental Results

A. Infrared Image Enhancement

We evaluate our method on infrared image enhancement using the HM-TIR dataset [10]. The visible-pretrained foundation model is equipped with task-responsive data generation and a thermal-aware lightweight adapter for infrared image enhancement. We compare our method with representative non-uniformity correction methods, including UTV [61], WFAF [62], GF [63], LRSID [64], SEID [65], ASCNet [66], and PPFN [10]. For infrared super-resolution, we further compare with CAT [67], CATA [68], CoRPLE [27], DAT [69], InfraFFN [70], IRSRMamba [71], OSEDiff [72], SwinIR [73], and TGDM [74]. All learning-based methods are fine-tuned on the HM-TIR dataset following their original settings.

Infrared Non-uniform Correction. Table I and Fig. 2 present the quantitative and qualitative comparisons under moderate and severe stripe degradations. As shown in Table I, our method achieves the best overall performance under both degradation levels. Under moderate degradation, our method achieves the best performance across all reported metrics. Under severe degradation, it obtains the best LPIPS, MUSIQ, CLIQA, and FID scores, while remaining competitive in terms of MANIQA. These results show that our method is effective in handling structured stripe interference across different image quality metrics. The visual results in Fig. 2 further confirm this advantage. Existing methods can partially suppress stripe noise but often leave residual stripes or introduce over-smoothed textures. In contrast, our method removes stripe artifacts more effectively while preserving object boundaries, human silhouettes, and background structures, producing clearer and more natural restored images.

Infrared Image Super-resolution. For infrared image super-resolution, we evaluate different methods under $\times 2$ and $\times 4$ scale factors, as reported in Table II. Our method achieves the best overall performance across both settings, indicating

TABLE II
QUANTITATIVE COMPARISON OF INFRARED IMAGE SUPER-RESOLUTION PERFORMANCE UNDER DIFFERENT SCALE FACTORS.

Degrees	×2 Scale					×4 Scale				
Methods	LPIPS↓	MUSIQ↑	CL-IQA↑	MANIQA↑	FID↓	LPIPS↓	MUSIQ↑	CL-IQA↑	MANIQA↑	FID↓
CAT	0.3666	24.21	0.1906	0.1692	83.84	0.4165	23.23	0.2123	0.1670	97.55
CATA	0.3553	24.13	0.1619	0.1672	81.73	0.4154	23.06	0.2133	0.1645	96.36
CoRPLE	0.3632	23.76	0.1636	0.1782	83.02	0.4155	23.61	0.1733	0.1810	97.66
DAT	0.3658	24.26	0.1857	0.1689	83.93	0.4166	23.15	0.2101	0.1659	97.21
InfraFFN	0.3339	24.15	0.1350	0.1717	79.37	0.3979	22.92	0.1578	0.1763	91.76
IRSRMamba	0.3569	24.13	0.1656	0.1774	81.40	0.4054	23.09	0.1486	0.1754	94.18
OSEDiff	0.3258	28.49	0.1565	0.1723	110.8	0.3391	32.40	0.1706	0.2005	115.3
SwinIR	0.2804	37.35	0.1751	0.2614	105.4	0.3501	35.58	0.1455	0.2407	149.7
TGDM	0.4132	25.42	0.1172	0.1724	126.1	0.4141	25.70	0.1663	0.1751	142.1
Ours	**0.1957**	**49.59**	**0.2735**	**0.2911**	**66.61**	**0.2111**	**49.90**	**0.2877**	**0.3008**	**71.97**

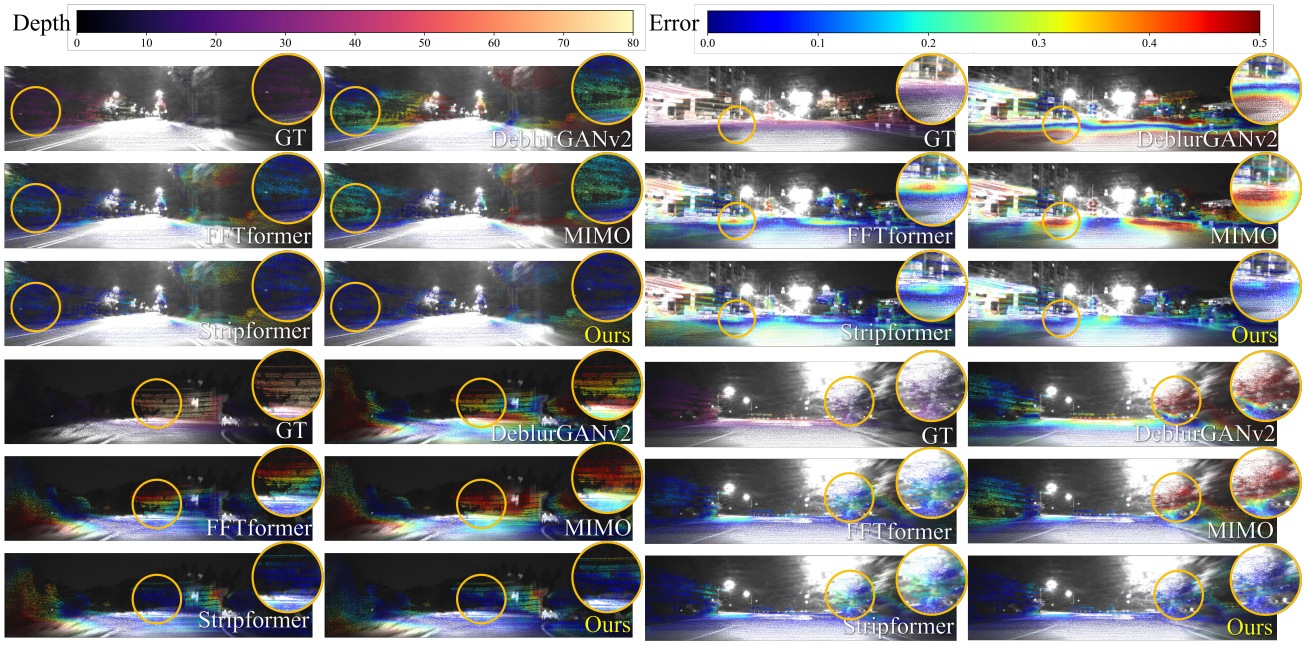


Fig. 4. Visual comparison of infrared depth estimation results on the MS2 dataset under motion blur degradation.

improved robustness to different degrees of resolution degradation. Compared with existing methods, which often suffer from performance degradation as the upsampling factor increases, our method better preserves high-frequency structures while maintaining perceptual quality and distributional consistency. The qualitative comparisons in Fig. 3 show similar trends. Existing methods can recover coarse image content but tend to produce blurred boundaries, over-smoothed textures, or artificial local patterns. In contrast, our method generates sharper and more coherent structures, with clearer object boundaries and more faithful local contrast. We further validate the proposed method on additional FLIR and TNO infrared images with real ×2 resolution degradation in Fig. 5. Compared with existing methods, our approach produces clearer structural boundaries and fewer restoration artifacts, while better preserving thermal details in the highlighted regions.

B. Infrared Depth Estimation

We evaluate infrared depth estimation under motion-blur degradation on the fixed MS2 subset described in Section IV and compare our method with representative restoration-based and infrared baselines, including MIMO [75], PPFN [10], FFTFormer [76], StripFormer [77], and DeblurGANv2 [78].

Quantitative Comparison. As shown in Table III, we report threshold accuracy and standard depth error metrics for comprehensive and fine-grained depth quality evaluation across multiple criteria. Our method achieves the best performance on most metrics, showing more accurate and robust geometric prediction under motion blur. In particular, the improvements in $\delta < 1.25$, Abs Rel, Log_{10}, RMSE log, SILog, and Sq Rel indicate more accurate depth prediction under motion blur. Although StripFormer performs slightly better in RMSE and the two relaxed threshold-accuracy metrics, our method achieves the best $\delta < 1.25$, Abs Rel, Log_{10}, RMSE log,

TABLE III
QUANTITATIVE COMPARISON OF INFRARED DEPTH ESTIMATION ON THE MS2 DATASET.

Methods	$\delta < 1.25 \uparrow$	$\delta < 1.25^2 \uparrow$	$\delta < 1.25^3 \uparrow$	Abs Rel↓	RMSE↓	$\text{Log}_{10} \downarrow$	RMSE log↓	SILog↓	Sq Rel↓
MIMO	0.6992	0.9135	0.9788	0.1669	5.5673	0.0807	0.2362	19.0581	1.1381
PPFN	0.8475	0.9860	0.9991	0.1301	3.5703	0.0536	0.1560	13.9171	0.5419
FFTFormer	0.8626	0.9906	0.9993	0.1215	3.5686	0.0512	0.1494	13.6735	0.5037
StripFormer	0.8751	**0.9948**	**0.9999**	0.1193	**3.4777**	0.0493	0.1422	12.6467	0.4819
DeblurGANv2	0.4077	0.7501	0.9182	0.2734	8.5281	0.1378	0.3817	30.6982	2.4623
Ours	**0.8849**	0.9937	0.9997	**0.1117**	3.6234	**0.0469**	**0.1384**	**12.3091**	**0.4776**

TABLE IV
QUANTITATIVE COMPARISON ON NUDT-SIRST, IRSTD-1K AND IRSTDID-SKY DATASETS.

Datasets	NUDT-SIRST				IRSTD-1k				IRSTDID-SKY				Overall			
Methods	mIoU↑	F1↑	P_d↑	F_a↓	mIoU↑	F1↑	P_d↑	F_a↓	mIoU↑	F1↑	P_d↑	F_a↓	mIoU↑	F1↑	P_d↑	F_a↓
ACM	64.85	78.68	96.72	0.0286	60.33	75.26	**93.27**	0.0685	44.30	61.40	93.47	0.0272	56.49	71.78	94.49	0.0414
ALCNet	61.13	75.88	97.25	0.0291	58.09	73.49	92.93	0.0745	26.15	41.46	98.43	0.0560	48.46	63.61	96.20	0.0532
MLCLNet	84.72	91.72	99.05	0.0102	63.62	77.76	91.58	0.0264	70.55	82.73	98.17	0.0461	72.96	84.07	96.27	0.0276
ISTDUNet	92.39	96.05	91.25	0.0301	63.90	77.98	91.25	0.0485	70.37	82.61	92.69	0.0264	75.55	85.55	91.73	0.0350
DNANet	94.19	97.01	99.26	0.0244	65.73	79.33	89.56	**0.0123**	71.66	83.49	96.34	0.0287	77.19	86.61	95.05	**0.0218**
RDIAN	82.42	90.36	98.84	0.0148	59.93	74.95	87.21	0.0333	69.39	81.93	96.61	0.0388	70.58	82.41	94.22	0.0290
UIUNet	90.25	95.02	98.84	0.0834	65.69	79.29	91.25	0.0135	72.03	83.74	96.61	0.0223	75.99	86.02	95.57	0.0397
SDSNet	94.38	94.02	98.62	0.0728	64.84	78.67	93.26	0.2535	70.24	82.51	96.61	0.0183	76.49	85.07	96.16	0.1149
SAM-SPL	94.63	97.24	**99.47**	0.0255	64.28	78.25	91.92	0.1438	71.01	83.04	96.79	**0.0057**	76.64	86.18	96.06	0.0583
Ours	**95.26**	**97.57**	99.26	**0.0064**	**66.40**	**79.81**	92.26	0.2019	**72.65**	**84.16**	**98.72**	0.0079	**78.10**	**87.18**	**96.75**	0.0743

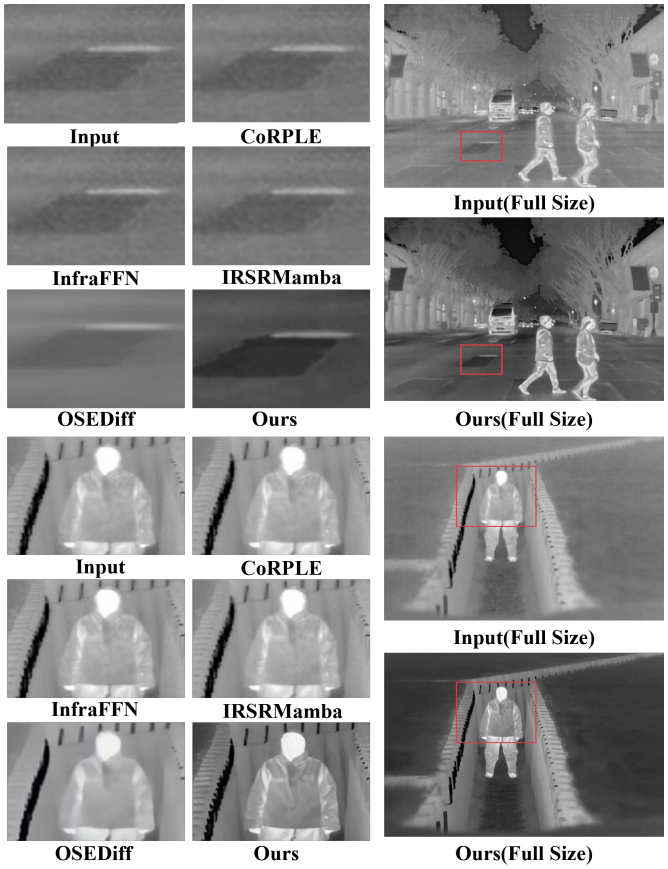


Fig. 5. Cross-dataset qualitative comparison on FLIR and TNO.

SILog, and Sq Rel scores. These results suggest that an explicit deblurring stage does not necessarily yield the best downstream depth estimation performance.

Qualitative Comparison. Fig. 4 shows visual comparisons of infrared depth estimation on the MS2 dataset under motion blur degradation, where blue indicates smaller errors and yellow or red denote larger errors. As highlighted by the yellow circles, motion blur severely affects distant regions, vegetation, low-contrast structures, and object boundaries. Compared with restoration-based pipelines such as DeblurGANv2, FFTFormer, StripFormer, and MIMO, our method produces more coherent depth maps with fewer local artifacts. The error maps further show that our results contain more blue and fewer yellow/red responses in challenging regions, indicating lower local prediction errors and better consistency.

C. Infrared Small Target Detection

We compare our method with representative methods, including ACM [30], ALCNet [79], MLCLNet [80], ISTDUNet [81], DNANet [6], RDIAN [82], UIUNet [31], SDSNet [83], and SAM-SPL [33]. We evaluate the proposed training strategy for infrared small target detection on NUDT-SIRST [6], IRSTD-1k [16], and IRSTDID-SKY [59]. Unless otherwise specified, all F_a values in infrared small target detection experiments are reported in units of 10^{-4}.

Quantitative Comparison. As shown in Table IV, our method consistently achieves the highest mIoU and F1-score across all three datasets. In the overall comparison, it improves mIoU, F1-score, and P_d by 0.91, 0.57, and 0.48 percentage points

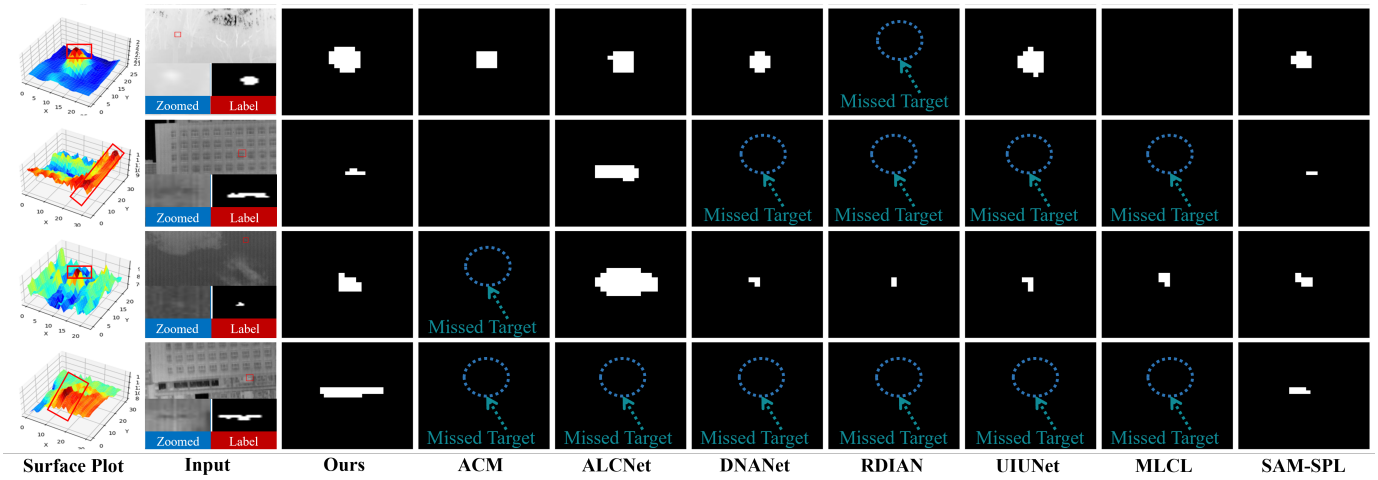


Fig. 6. Qualitative comparison of infrared small-target detection on NUDT-SIRST, IRSTD-1k, and IRSTDID-SKY.

TABLE V
QUANTITATIVE SEMANTIC SEGMENTATION PERFORMANCE COMPARISON UNDER SUPER-RESOLUTION WITH DIFFERENT SCALING FACTORS.

Scale	Methods	Unlab.	Road	Sd.walk	Build.	T.Light	T.Sign	Veg.	Sky	Pers.	Car	Truck	Bus	Motor.	Pole	mIoU↑
×2	CATA	13.80	87.29	39.15	78.59	18.26	59.43	80.92	92.62	60.08	76.42	35.43	13.94	30.73	35.45	51.58
	DAT	13.78	87.29	39.16	78.56	18.29	59.46	80.90	92.64	60.15	**76.43**	35.36	13.94	31.16	35.45	51.61
	CAT	13.78	87.30	39.14	78.56	18.30	59.45	80.90	92.63	60.17	76.42	35.33	13.93	**31.37**	35.43	51.62
	CoRPLE	13.73	87.28	39.13	78.57	18.92	**59.49**	80.89	92.63	60.09	76.42	**35.51**	13.98	30.91	35.41	51.64
	SwinIR	13.77	87.29	39.16	78.57	18.25	59.44	80.91	92.63	60.12	76.42	35.45	13.92	30.84	**35.49**	51.59
	Ours	**16.05**	**87.85**	**40.97**	**78.88**	**23.16**	58.59	**81.08**	**93.01**	**64.72**	75.19	29.72	**15.75**	28.94	34.50	**52.03**
×4	CATA	13.55	87.20	38.78	78.58	18.92	58.91	80.65	92.69	59.76	75.92	22.09	12.17	27.97	34.92	50.15
	DAT	13.67	87.23	38.93	**78.73**	18.73	58.65	**80.76**	92.71	60.26	**76.02**	21.85	**13.01**	29.73	**35.05**	50.38
	CAT	13.60	87.22	38.83	78.69	19.31	58.68	80.72	92.72	60.15	76.01	22.28	12.92	**29.75**	35.02	50.42
	CoRPLE	13.59	87.19	38.84	**78.73**	18.27	58.95	80.67	92.74	60.15	75.98	22.53	12.59	28.57	34.97	50.27
	SwinIR	13.58	87.22	38.78	78.64	19.56	58.87	80.69	92.68	59.99	75.95	22.12	12.84	28.57	34.96	50.32
	Ours	**15.95**	**87.42**	**40.32**	78.39	**27.81**	**60.09**	79.94	**92.83**	**63.47**	72.47	**30.18**	12.17	22.21	32.60	**51.13**

over the strongest competing results, respectively. On NUDT-SIRST, our method improves the best competing mIoU and F1-score by 0.63 and 0.33 percentage points, respectively, while reducing F_a from 0.0102 to 0.0064. On IRSTD-1k, it improves the best competing mIoU and F1-score by 0.67 and 0.48 percentage points. On IRSTDID-SKY, the corresponding improvements in mIoU, F1-score, and P_d are 0.62, 0.42, and 0.29 percentage points, respectively. Although several competing methods achieve lower F_a on specific datasets, our method provides stronger region-level prediction accuracy, as reflected by its consistently higher mIoU and F1-score.

Qualitative Comparison. Fig. 6 presents visual comparisons on NUDT-SIRST, IRSTD-1k, and IRSTDID-SKY. The selected examples contain weak targets, low contrast, background clutter, and noise-like interference, which are common challenges in infrared small target detection. As observed, many existing methods either miss dim targets or produce incomplete masks when the target response is weak. Compared with these methods, our method generates more stable and complete predictions. It better preserves small target structures while suppressing irrelevant background responses. In cluttered regions, competing methods may confuse bright background edges or noise-like regions with true targets, leading to false alarms or fragmented masks. By contrast, our method produces compact and accurate target masks that are more consistent with the ground truth. These visual results are consistent with the quantitative improvements in Table IV, demonstrating the robustness of the proposed method across all three datasets under challenging infrared scenes.

D. Infrared Semantic Segmentation

Quantitative Comparison. We first compare different methods under stripe noise corruption with two severity levels, as shown in Table VI. Our method achieves the best overall mIoU across all degradation levels and maintains stable performance as the stripe noise becomes stronger. This indicates that our method better preserves semantic structures and provides more reliable segmentation under structured interference. We further compare our method with super-resolution-based segmentation pipelines under $\times 2$ and $\times 4$ resolution degradation, as reported in Table V. Unlike existing methods that first restore low-resolution inputs using blind SR models, our method directly operates on interpolated inputs without explicit reconstruction. Despite this simpler pipeline, it achieves the best overall performance under both degradation settings, suggesting that

TABLE VI
PERFORMANCE COMPARISON OF SEMANTIC SEGMENTATION UNDER STRIPE NOISE WITH DIFFERENT SEVERITY LEVELS.

Severity	Methods	Unlab.	Road	Sd-walk	Build.	T-Light	T-Sign	Veg.	Sky	Pers.	Car	Truck	Bus	Motor.	Pole	mIoU↑
Mod.	ASCNet	15.01	84.66	34.92	69.24	22.71	38.20	69.24	89.59	58.39	75.38	32.72	11.41	25.82	28.32	46.83
	LRSID	15.53	85.45	36.84	72.00	26.23	46.03	72.27	90.50	58.81	75.61	34.92	10.40	27.58	29.74	48.71
	WFAF	15.26	86.73	38.50	74.49	19.83	53.22	76.30	91.63	**58.94**	76.09	34.66	11.30	29.34	30.84	49.80
	GF	14.32	83.22	33.45	67.79	**29.90**	34.37	66.85	89.23	54.46	72.68	26.74	6.11	17.71	24.17	41.40
	HM-TIR	14.23	86.63	37.65	75.33	11.76	54.66	77.71	90.69	57.77	75.95	**35.13**	10.43	30.00	31.26	49.23
	Ours	**16.41**	**87.66**	**41.19**	**77.72**	16.96	**58.04**	**80.41**	**92.88**	56.88	**77.03**	21.64	**15.27**	**35.87**	**35.05**	**50.93**
Sev.	ASCNet	13.95	84.30	34.22	70.27	16.65	38.17	69.81	90.51	58.35	75.57	28.07	13.28	27.54	29.55	46.45
	LRSID	13.77	83.37	33.82	50.56	20.14	34.31	64.19	67.39	57.71	72.96	29.41	10.58	27.31	22.76	42.02
	WFAF	14.78	86.34	37.15	67.43	21.92	50.04	73.80	85.84	58.73	75.41	33.31	13.40	29.97	27.57	48.26
	GF	13.27	83.14	32.29	57.86	**23.23**	34.06	64.04	78.74	54.02	71.91	25.32	6.66	18.80	21.28	41.76
	HM-TIR	14.03	86.50	38.49	74.96	14.45	54.23	78.22	89.53	**59.05**	75.43	**34.05**	14.25	**32.80**	26.85	49.49
	Ours	**16.32**	**87.69**	**41.46**	**77.42**	16.65	**58.05**	**80.29**	**92.74**	57.04	**77.25**	21.16	**15.06**	31.83	**34.89**	**50.56**

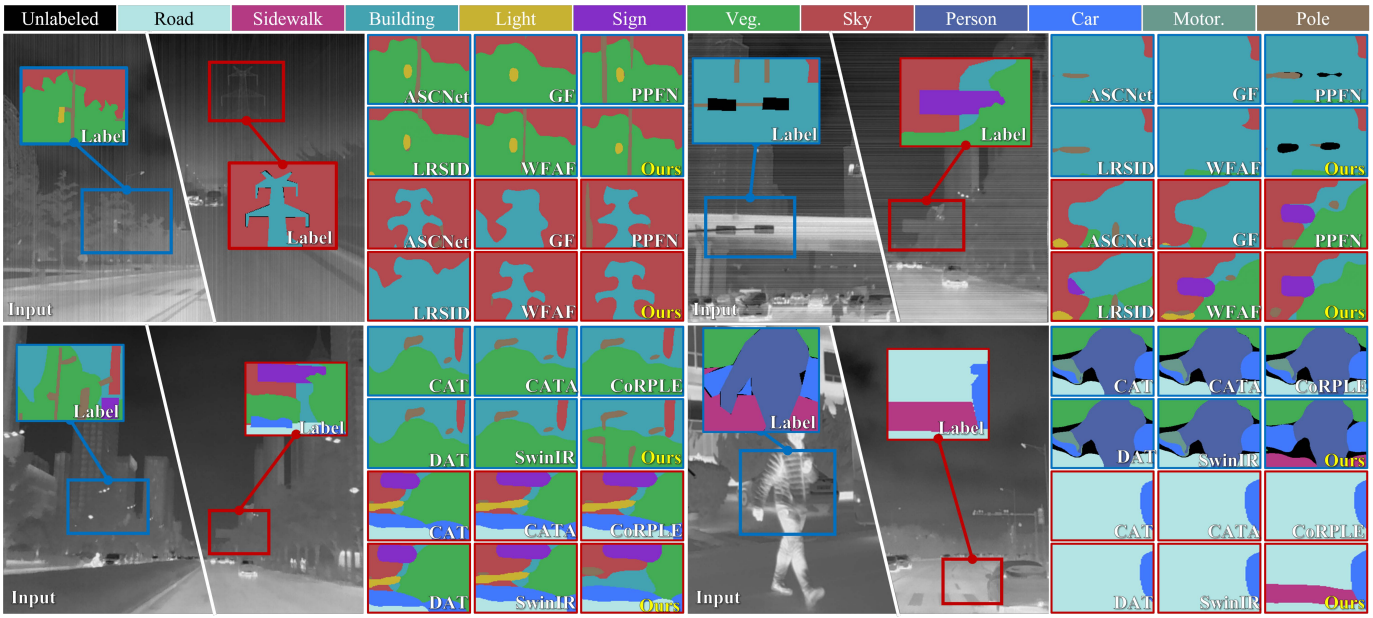


Fig. 7. Visual comparison of infrared semantic segmentation results on the FMB dataset.

avoiding error-prone SR reconstruction helps preserve discriminative semantic cues in degraded infrared scenes.

Qualitative Comparison. Fig. 7 shows visual comparisons of infrared semantic segmentation on the FMB dataset under stripe noise and resolution degradation. Under stripe noise, competing methods often miss object regions or produce fragmented segmentation masks, especially for small objects, thin structures, and low-contrast class boundaries. The stripe artifacts introduce strong structured interference, making it difficult to distinguish semantic foreground regions from background patterns. In contrast, our method preserves more complete class regions and clearer object boundaries, indicating stronger robustness against stripe-induced feature disturbance. Under resolution degradation, competing methods tend to suffer from boundary ambiguity and category-level inconsistency, leading to incomplete object masks and inaccurate region predictions after reconstruction. Our method maintains more reliable semantic regions and sharper contours under both $\times 2$ and $\times 4$ settings, demonstrating that thermal-aware model transfer helps preserve task-relevant semantic cues for dense prediction. These results suggest that direct prediction from degraded inputs better retains task-relevant semantic cues than the evaluated restoration-based pipelines.

E. Ablation Studies

We conduct diagnostic experiments to analyze four aspects of the proposed framework: training-data efficiency, task-responsive data generation, thermal-aware model transfer, and single-loop collaborative optimization. We first examine the sensitivity of infrared restoration performance to different training-data proportions. We then use infrared small-target detection as a representative diagnostic task because it is highly sensitive to both imaging variations and model transfer.

Data-Efficiency Analysis on Restoration. Table VII evaluates the sensitivity of the full framework to different training-data proportions. Under moderate degradation, using 80% of

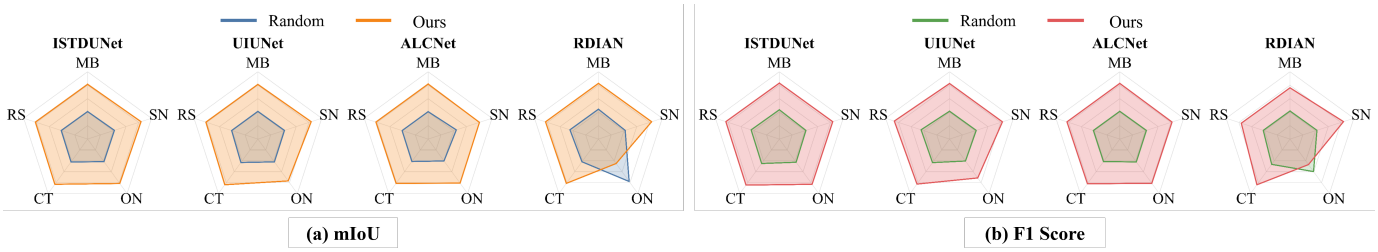


Fig. 8. Radar-chart comparison between random augmentation and task-responsive data generation under five infrared imaging variations: Motion Blur (MB), Stripe Noise (SN), Optical Noise (ON), Contrast variation (CT), and Resolution variation (RS).

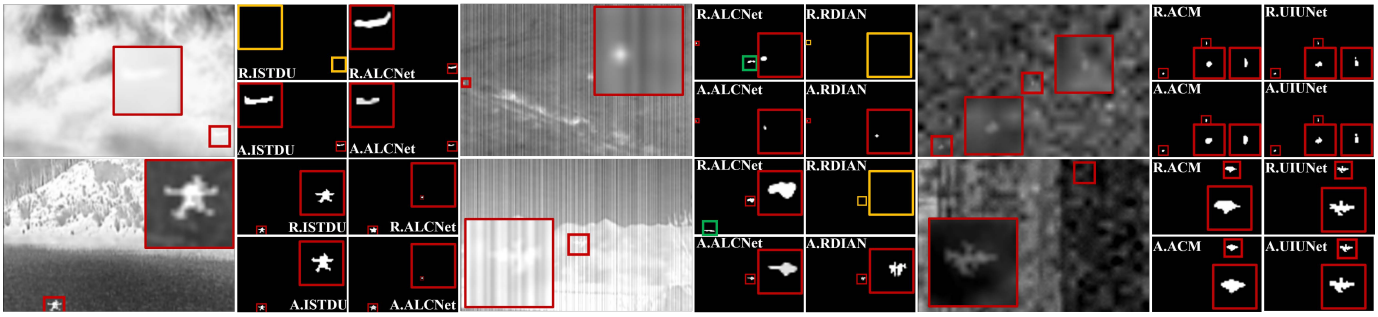


Fig. 9. Visual comparison between random augmentation and the proposed task-responsive data generation under three representative imaging variations: (a) motion blur, (b) stripe noise, and (c) optical noise. "A." denotes task-responsive data generation and "R." denotes random augmentation.

TABLE VII
ABLATION STUDY ON TRAINING DATA PROPORTIONS.

Degradation	Ratio	PSNR↑	SSIM↑	LPIPS↓	FSIM↑
Moderate	20%	25.8432	0.8305	0.1343	0.9240
	40%	26.4445	0.8359	0.1233	0.9301
	60%	28.0498	0.8480	0.1205	0.9325
	80%	**28.5544**	**0.8613**	0.1106	**0.9405**
	100%	27.9740	0.8514	**0.1087**	0.9323
Severe	20%	24.5556	0.7419	0.2552	0.8471
	40%	24.8938	0.7535	0.2505	0.8520
	60%	25.4787	0.7573	0.2427	0.8572
	80%	25.3536	0.7616	0.2810	0.8489
	100%	**26.1593**	**0.7666**	**0.2020**	**0.8627**

the training data achieves the best PSNR, SSIM, and FSIM, while the full set gives the lowest LPIPS. Under severe degradation, the full set performs best overall, but the 60%-80% settings remain competitive. These results indicate that the full framework retains competitive restoration performance when the amount of training data is moderately reduced.

Effectiveness of Data Generation. Fig. 8 and Fig. 9 evaluate the proposed task-responsive data generation against conventional random augmentation for infrared small target detection. The comparison is conducted under multiple degradation types, including motion blur, stripe noise, optical noise, contrast degradation, and resolution degradation, and covers multiple detector architectures. As shown in Fig. 9, random augmentation can expose the detector to corrupted inputs, but its degradation sampling is independent of the detector's current failure modes. Consequently, the resulting predictions remain sensitive to background clutter and structured interference. For example, under motion blur, the target responses become spatially diffused, leading to incomplete or imprecise masks. In contrast, task-responsive data generation produces more compact and complete target masks across these degradation types, suggesting that the generated samples effectively guide the model to learn more robust target-background discrimination. The radar plots in Fig. 8 provide a quantitative comparison between random and task-responsive data generation in terms of mIoU and F1. Each axis corresponds to a specific degradation type, and the comparison is performed across representative detector architectures. Compared with random augmentation, task-responsive data generation generally expands the radar area for both mIoU and F1, indicating more stable segmentation quality under diverse corruptions. The improvement is particularly evident for degradation types that strongly affect small-target saliency, such as stripe noise and motion blur. Moreover, the consistent gains across different detector architectures suggest that the benefit of task-responsive generation is not tied to a specific detector architecture, but reflects a general training effect that improves thermal-aware robustness. Fig. 10 visualizes the task-responsive imaging-variation weights predicted by the generator over training iterations. The response maps reveal distinct preferences over different infrared imaging variations. For the point-object scene, the generator assigns higher weights to stripe noise and contrast degradation, indicating that weak point targets are more vulnerable to the structured interference. For the UAV-object scene, higher responses are observed for optical noise and motion blur, suggesting that larger targets with shape cues are more affected by boundary disturbance and blur-induced structural distortion. These dynamic patterns

indicate that the generator adjusts its imaging-variation preferences according to the input characteristics and training status.

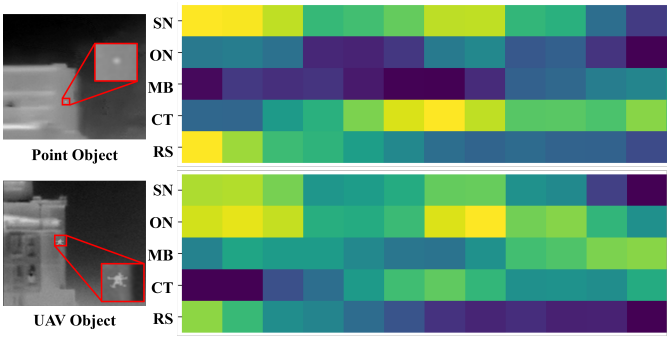


Fig. 10. Visualization of task-responsive imaging-variation weights for different infrared scenes.

TABLE VIII
QUANTITATIVE COMPARISON OF DIFFERENT PEFT STRATEGIES ON NUDT-SIRST AND IRSTDID-SKY DATASETS.

Method	Params(M)	NUDT-SIRST		IRSTDID-SKY	
		mIoU↑	F1↑	mIoU↑	F1↑
Full FT	25.60	94.65	97.25	51.43	67.93
LoRA	0.1537	94.39	97.11	71.72	83.53
Q-LoRA	0.1542	94.66	97.26	71.64	83.47
Adapter	0.4218	94.88	97.37	70.12	82.44
DoRA	0.1626	94.69	97.27	71.05	83.07
Bi-LoRA	0.1994	93.07	96.41	71.21	83.19
SSF	0.0778	94.44	97.14	70.12	82.44
Rein	3.4027	94.94	97.41	71.89	83.65
Ours	0.0946	**95.26**	**97.57**	**72.65**	**84.16**

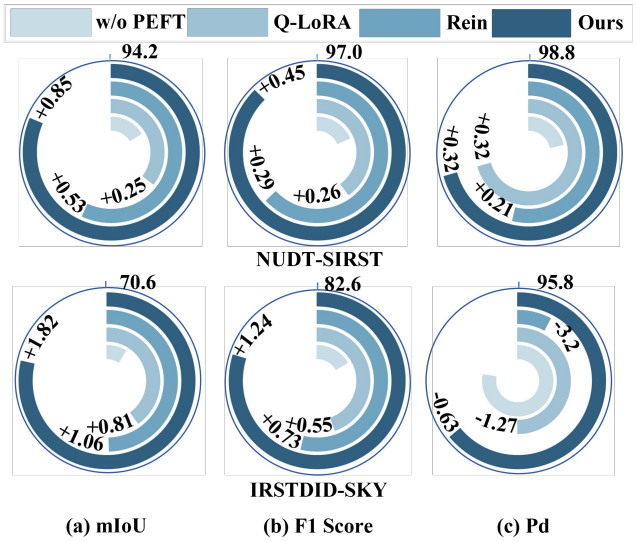


Fig. 11. Comparison with PEFT strategies on two IRSTD datasets.

Effectiveness of Model Transfer. Table VIII, together with Fig. 11, shows that the proposed model-transfer strategy achieves a favorable balance between detection accuracy and parameter efficiency. Although generic PEFT schemes, including LoRA [21], Q-LoRA [84], DoRA [22], Adapter [24], SSF [85] and Rein [48], introduce lightweight updates to frozen foundation models, their update mechanisms are largely modality-agnostic and do not explicitly exploit the formation characteristics of infrared observations. In contrast, our method achieves the best mIoU and F1-score on both datasets while using fewer trainable parameters than most competing methods. Compared with full fine-tuning, our method achieves higher accuracy with approximately 0.37% of its trainable parameter budget, suggesting that updating the entire backbone is not necessarily beneficial under limited infrared supervision and may increase the risk of overfitting. The advantage is particularly evident compared with Rein, where our method achieves higher detection accuracy using only a small fraction of its trainable parameters. These results suggest that the performance advantage is primarily associated with infrared-specific conditioning rather than simply increasing the number of trainable parameters.

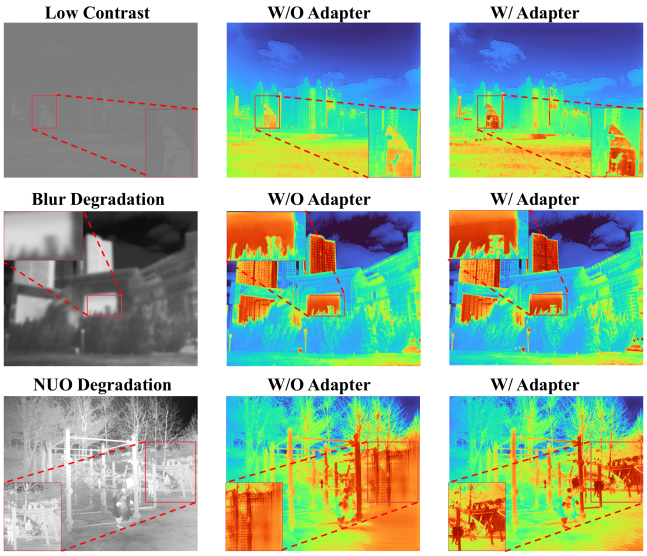


Fig. 12. Visualization of feature responses with and without the proposed adapter under different infrared degradations.

TABLE IX
ABLATION STUDY OF ADAPTER COMPONENTS.

Variant	mIoU↑	F1↑	P_d↑	F_a↓
w/o FFT	72.01	83.72	96.79	0.0172
w/o Modulation	69.94	82.31	98.71	0.0398
w/o Guidance	70.67	82.82	98.71	0.0208
Full	**72.65**	**84.16**	**98.72**	**0.0079**

Fig. 13 visualizes the complementary responses of the structure and detail features in two infrared scenes. In the upper scene, the small target has a weak response against a textured background. The response-detail map retains the localized target response highlighted by the red box and captures fine intensity variations in the surrounding background. The radiance-structure map provides a smoother description of the background regions, emphasizing their spatial organization

rather than individual texture responses. In the lower urban scene, the response-detail map emphasizes building edges, pedestrian contours, and local variations around vehicles. The radiance-structure map captures broader building regions and more continuous foreground responses, providing a clearer representation of the scene layout. These observations illustrate the different roles of the two features: the structure component supplies spatial context, while the detail component retains localized cues that may be weakened in a coarse representation. Their combination provides complementary guidance for the thermal-aware adapter, allowing model transfer to use both region-level structure and fine-grained infrared responses.

TABLE X
QUANTITATIVE COMPARISON OF DIFFERENT TRAINING STRATEGIES ON NUDT-SIRST AND IRSTDID-SKY.

Strategy	NUDT-SIRST		IRSTDID-SKY	
	mIoU↑	F1↑	mIoU↑	F1↑
Alternative	93.74	96.77	70.65	82.80
End-to-End	94.46	97.15	70.71	82.84
Stage-wise	91.65	95.64	65.99	79.51
Ours	**95.26**	**97.57**	**72.65**	**84.16**

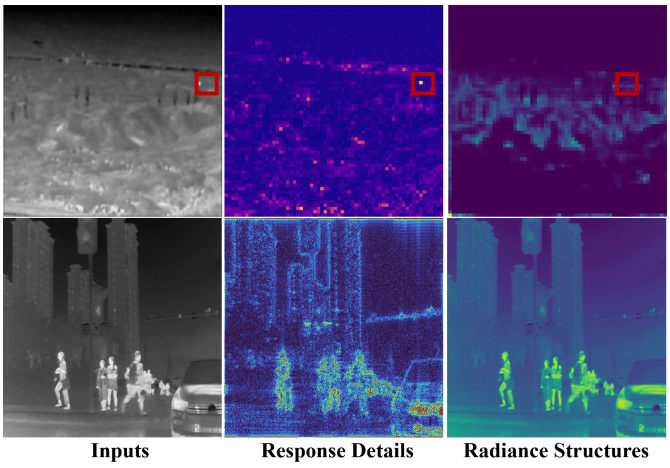


Fig. 13. Visualization of response-detail and radiance-structure features in different infrared scenes.

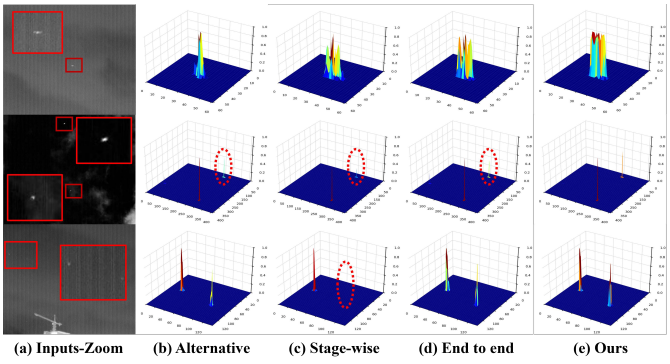
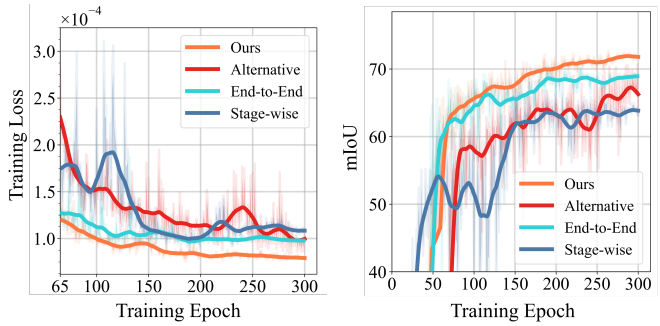


Fig. 14. Visualization of target detection under different training strategies.

Fig. 12 and Table IX further validate the effectiveness of the proposed adapter. As shown in Fig. 12, without the adapter, feature responses are easily disturbed by low contrast, blur, and non-uniform noise, resulting in weakened structural cues and confused background activations. With the adapter, the responses become more concentrated on semantically meaningful regions and preserve clearer object boundaries, indicating that the adapter can enhance structure-relevant features while suppressing degradation-induced interference. The quantitative ablation in Table IX shows consistent trends. Removing the frequency transformation branch causes a clear performance drop, suggesting the importance of spectral cues for weak target enhancement. Removing the modulation leads to a larger drop in mIoU and F_1 and increases the false-alarm rate, demonstrating the importance of descriptor-conditioned feature modulation. The variant without guidance also underperforms the full model, demonstrating that the proposed guidance provides useful structure-detail cues for IRSTD.



(a) Train Loss (b) Validation Performance

Fig. 15. Training dynamics of different training strategies.

Effectiveness of Single-Loop Optimization. Table X compares four strategies for training the representation module and thermal-aware adapter on NUDT-SIRST and IRSTDID-SKY, using mIoU and F1-score as evaluation metrics. All strategies use the same generator-update procedure. Alternative strategy updates the adapter and representation module in turn, keeping the other component fixed at each step. End-to-end strategy updates both components jointly. Stage-wise strategy first trains the adapter with the representation module fixed, then freezes the adapter and trains the representation module. Our method jointly updates both components through the proposed single-loop objective, which accounts for how the learned representation affects adapter optimization. Our method achieves the highest scores on both datasets: 95.26% mIoU and 97.57% F1 on NUDT-SIRST, and 72.65% mIoU and 84.16% F1 on IRSTDID-SKY. Compared with End-to-end strategy, the strongest baseline, the mIoU gains are 0.80 and 1.94 percentage points, respectively. Stage-wise strategy gives the lowest scores on both datasets, suggesting that training the two components separately limits their ability to adapt to each other. End-to-end strategy performs better than Alternative, but remains below our method. These results support learning the representation together with the adapter it guides, and show that the proposed single-loop strategy coordinates their

learning more effectively than the compared alternatives.

Fig. 14 illustrates the small-target detection behavior under different training strategies. The alternative and end-to-end strategies can activate target-related regions, but they also produce dispersed responses and background false peaks, indicating insufficient discrimination between weak targets and confusing background structures. The stage-wise strategy reduces some background interference, yet it tends to suppress weak target responses and may lead to missed detections, as highlighted by the circled regions. In contrast, the proposed strategy produces sharper and more compact response peaks at true target locations while maintaining low activation in surrounding background areas. Fig. 15 further analyzes the training dynamics of different strategies. Compared with the alternative, stage-wise, and end-to-end schemes, our method achieves a smoother and lower training loss, indicating that the coordination mechanism mitigates the optimization oscillations caused by the interaction between task-responsive data generation and model transfer. Meanwhile, the validation mIoU of our method increases more rapidly and converges to a higher level, whereas the competing strategies exhibit larger fluctuations, slower convergence, or earlier saturation. These results suggest that the proposed single-loop strategy better coordinates the coupled data- and model-side objectives, thereby improving optimization stability.

VI. Conclusions

This paper presents a data-model collaborative transfer framework for infrared visual perception. At the data level, the framework exploits infrared imaging regularities to construct a feasible imaging-variation space and learns task-responsive weights to synthesize informative training observations. At the model level, it derives radiance-structure and response-detail descriptors through a formation-guided proxy representation, which conditions lightweight adapters for thermal-aware model transfer. A single-loop collaborative optimization strategy is further introduced to coordinate data generation, representation learning, and model transfer without expensive inner-loop convergence. Extensive experiments across four representative infrared perception tasks demonstrate the broad applicability and parameter efficiency.

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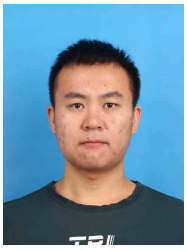

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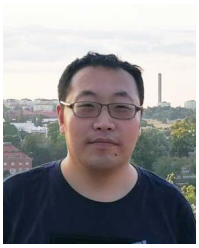

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